FACULTY OF APPLIED SCIENCE
RAJARATA UNIVERSITY OF SRI LANKA

ROAD DAMAGE DETECTION PROJECT

IMAGE ACQUISITION

2403 - IMAGE PROCESSING

LECTURER - DR. K.A.S.H. KULATHILAKE

M.I.M ARAFATH - ICT/2023/142 - 6119

K. IFFATH - ICT/2023/145 - 6121

B.G.N.C - ICT/2023/141 - 6118

M.I.F ILMA - ICT/2023/146 - 6336

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1. Video Capture Device

This video was shot from a rear camera of the mobile phone, specifically the **iPhone 14 Pro.** In shooting this video, the setting of the 1× wide-angle rear camera was used for shooting this video, and the other **two rear cameras** were **not used**. The wide-angle camera has contained a **26 mm equivalent lens**, with **an f/1.5 aperture** and a **1.9 µm pixel** size to let in more light sensitivity for better image clarity. It also features **sensor-shift image stabilisation (OIS)** and dual-pixel phase detection autofocus for stable and sharp video captures even with the minor camera movements.

The frame rate of a mobile phone can be as **low as 24 FPS** and as **high as 60 FPS**, thus allowing flexibility between requirements of motion smoothness. Besides that, various resolutions can be used: **HD, Full HD,** and **4K**, meaning that this device is highly suitable for high-quality video acquisition tasks.

2. Recording Parameters and Their Significance

a) Resolution

In this project the video used has been recorded in **Full HD** resolution of **1920 x 1080 pixels**. The chosen resolution can record an ideal balance of details of an image and processing power of the central processing unit. Full HD resolution is capable of **detecting minute details** like cracks, potholes, and transverse cracking on the image or video surfaces. Such details are critical in accurate image or video processing analysis. In addition, when considered in relation to other high resolutions like 4K media, Full HD resolution is significantly faster in its processing performance.

b) Frame Rate

The frames per second of our recorded videos are **30 FPS**. This ensures that there are no gaps in the recording of the road surface while moving. This enables to cracks, potholes, or road surface irregularities to be visible prominently in the images. By using a frame rate of 30 FPS, it is possible to **avoid motion blur** and the **occurrence of holes** in the frames. On the other hand, it maintains a balance between the **accuracy of detection** and the **processing time**. It is because a frame rate of **30 FPS will not increase processing time unnecessarily**.

c) Duration of Video

The video clips were recorded with a duration of 31 seconds. The recorded video includes several road damages such as cracks, potholes, alligator cracking, and ravelling. This duration video **reduce file size** and **processing time**, therefore making frame-by-frame analysis **more efficient**.

3. Environmental Conditions

The recording of the video was done in natural **outdoor conditions**; special attention was given to environmental factors since they directly influence the quality of images. Video capture was generally done in the **mornings** but extended in several cases until around **3:00 PM**. The videos were recorded under **natural daylight** conditions, but because of intermittent **cloudy and rainy** conditions, lighting intensity was subject to change.

The recording period coincided with **rainy days,** and as such some of the road surfaces **were wet**. In some places the **potholes** were **partially filled with rainwater** distorting their appearance.

Although such difficulties were faced, these videos can still be useful to analyse because they reflect realistic environmental conditions encountered in practical road surface inspection scenarios.

4. Camera Motion and Placement

The camera had an angled **position** of approximately **25° to 40°** with its lens directed towards the road surface angled **downwards**. The angle was selected to record the road extensively as well as its details. The camera was placed **1 to 1.5 metres** above the road surface. This was the best position as the footage captured an equal view with little distortion. The video recording is done in a handheld manner when the recorder is walking. There is also **no tripod or vehicle mounting** in place.

5. Site Selection

The selected road segments are **rural carpet roads**. Some of the selected road sections are where we captured these types of visible surface damages such as **Potholes**, **edge cracking**, **Alligator cracking and Revelling**. We believe these types of damages provide contrastingly different visual pattern variations, which are very useful in the process of image processing and analysis.

6. Recording Process and Sample Video

Recorded one video based on a systematic video recording scheme. First, appropriate rural road sections were shortlisted on the basis of observable surface conditions and road safety. **Before commencing** with the recordings, **traffic** and **people flow** were observed for a safe environment. It was ensured that no vehicle was approaching, and a **smooth walking pattern** was maintained throughout the video recording process. Additionally, video recordings **contained excessive camera shaking because we recorded video while walking so the small up-and-down motion is a result of the natural body motion while walking while taking the video**. The recording was immediately stopped in instances where vehicles suddenly appeared during the recording for **safety reasons**, and such **video clips were removed** from the dataset. vehicle or people obstructions, or any **unsafe video** recording conditions were **eliminated**, and only smooth video recording were considered for analysis purposes.

7. Video Analysis and Dataset Preparation

The recorded videos were then organised for processing. We used an iPhone 14 Pro, which meant that the **default video file** was in the **MOV** format. which stores high-quality visual information. The following step will be the frame extraction process, where each recorded video was processed individually to obtain frames (images). The process will be carried out using tool OpenCV for Python programming: The sampling process took place after the extraction of the frames, where every second we get one frame will be considered for extraction. The reason for the selection process was easy to testing and processing simplicity. After finished our testing we extract all frames.

8. Data Annotation Process

Annotation plays a very crucial part in the evaluation process, as it gives us the ground-truth labels needed in determining the effectiveness of our road damage detection approach. In this project, data annotation is done manually by highlighting the damaged parts like cracks or potholes in the images using the tool **Microsoft Paint**.

For each captured road image frame, the damaged sections are drawn by the user in order to mark and differentiate their ROI. This is done using the annotation tool's features for drawing lines, shapes, and other elements.

After marking the required areas in the images, they are processed to form binary images which contain pixels labeled '1' for representing damaged sections and '0' for undamaged road sections.

Ground Truth binary masks are then created for comparison against the output of the proposed road damage detection algorithm, in order to calculate evaluation parameters like Precision, Recall, F1-Score, and Intersection Over Union (IoU).

9. Ethical and Safety Considerations

The process of acquiring the videos was done to ensure that ethical aspects are considered. For instance, during the recording of the videos on the road surface condition, great caution was taken to **ensure** that **no identifiable person was captured during the recording.** This was because identification of a person was considered **not good** in the study because the key concern was the road surface. In terms of **safety**, it should be noted that the video recording system was subjected to the highest level of priority or not. We checked. Video recording was done only when **road traffic conditions were observed**, and it was **immediately halted** whenever vehicles and people came into sight. It is very clear that video acquisition was done with a high level of attention regarding **ethical responsibilities** and **safety** measures.